

SEMI-CAVA: A Causal Variational Approach to Semi-Supervised Learning

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Abstract—Deep learning has advanced rapidly, but relies heavily on large-labeled datasets for effective training. This is particularly challenging in fields like medicine, where expert labeling is costly, labor-intensive, and prone to bias and error. Semi-supervised learning (SSL) addresses this challenge by reducing reliance on labeled data. SSL is closely tied to the concept of causation. However, recent works relating causality to SSL are limited by modeling only low-dimensional observations or designing a plug-in module to alleviate the class imbalance. In this paper, we take steps towards training causal generative models for semi-supervised learning, combining principles from causality and variational inference. We interpret the Mixup strategy as a stochastic intervention and introduce a consistency loss to promote coherent latent representations. Under reasonable assumptions, we provide theoretical guarantees that the learned latent representations align with true causal factors up to permissible ambiguities. The experimental results show the proposed approach achieves state-of-the-art performance on several medical datasets of different modalities. Additionally, we test our model on standard benchmarking datasets: CIFAR10, CIFAR100, and SVHN, where it achieves competitive performance.

Index Terms—Semi-supervised learning, causality, variational inference, MedMNIST.

I. INTRODUCTION

MOST deep learning (DL) models are trained via supervised learning, which requires a substantial labeled dataset. While it may be simple to gather a large amount of raw data, in many cases, it may be prohibitively expensive (for example, when the labeling needs to be done by a professional) or otherwise impractical to obtain the corresponding labels. Additionally, data labels might include confidential information. Semi-supervised learning (SSL) aims to significantly reduce the need for labeled data by allowing models to make use of unlabeled data during training.

SSL is intrinsically linked to the concept of causation. Schölkopf et al. [1] have conjoined SSL to the underlying

causal structure of the given learning problem. In particular, they proclaim that SSL should not be viable when predicting a target variable based on its causes (also known as causal learning), but it should be possible to predict a target variable from its effects (known as anticausal learning). von Kügelgen et al. [2] further extend the connection by considering the problem of predicting a target variable based on both its causes and its effects. Recent works on relating causality to SSL are limited by modelling only low-dimensional observations [1], [2] or designing a plug-in module to alleviate class imbalance [3]. [4] introduces a causal diagram to provide insight into the effectiveness of mainstream semi-supervised segmentation methods from a causal standpoint. These strategies further depend on the absence of hidden endogenous factors in the causal model. In this article, we take a variational inference approach to address these challenges.

Prior efforts have incorporated causality into variational methods for generating counterfactuals [5], addressing robustness issues [6], learning disentangled representation [7], causal effect inference [8] etc. However, in the realm of SSL, these strategies remain unexplored to their full potential. We aim to bridge this gap with the following contributions:

- We propose a deep generative model (causal VAE) that incorporates causal relationships into the design of the model to bolster semi-supervised learning. It accounts for possible interventions on the observed variables in the data generation process. We derive evidence lower bound (ELBO) losses that integrate the underlying causal structure for SSL.
- We describe the mixup [9] strategy as a stochastic intervention applied to the causal VAE system. This intervention, coupled with a proposed consistency loss, aims to construct a coherent latent space. Under reasonable assumptions on the generative process and the inference model, we theoretically guarantee that our learned representations align with the true latent factors, up to permissible ambiguities. These findings offer new causal insights into mixup techniques.
- Our proposed method achieves state-of-the-art semi-supervised classification on seven medical datasets across different modalities, using only 10% labeled data. It also demonstrates competitive results on benchmark datasets: CIFAR10, CIFAR100, and SVHN.

II. RELATED WORKS

The most effective semi-supervised learning approaches are based on two basic ideas: consistency regularization and

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pseudo-labeling. *Pseudo-labeling* produces an artificial label for unlabeled images and trains the model to predict the artificial label. *Consistency regularization* utilizes the principle that a classifier should return the same distribution of classes for an unlabeled sample even when the input sample has been perturbed. Although some research perturbs adversarially [10] or by utilizing mixup [11] or by employing dropout [12], [13], recent methods [14], [15], [16], [17] have achieved greater accuracy using domain-specific, sophisticated strong augmentations such as RandAugment [18], CTAugment [15], and CutOut [19]. In addition to employing strong augmentations for achieving high performance, ReMixMatch [15] and SelfMatch [20] integrate self-supervised learning modules; SimMatch [17] utilizes a memory buffer to keep the embeddings for labeled examples; FlexMatch [21] adopts a curriculum learning approach to dynamically leverage unlabeled data. In this work, we do not aim to compete with or improve upon such sophisticated methods. Instead, our main objective is to develop a causality-inspired variational framework for SSL. We use mixup and standard image augmentations (e.g., random flips and crops) to enforce consistency in the latent space.

Variational Autoencoders have found their ground in SSL with the seminal work by Kingma et al. [22]. They propose a “hybrid continuous-discrete mixture model where the different mixture components share parameters”, known as M2. In addition to that, a more straightforward generative classifier model M1 is developed with a focus on dimensionality reduction. This just learns a VAE across the entire data and then trains a classifier using the learned latent space. Furthermore, incorporating M1 and M2, they build a hybrid model (M1+M2) that uses representation learned from ELBO to infer labels. Ilse et al. [23] and Louizos et al. [24] incorporate domain-specific knowledge into models so that ELBO representations are pertinent to label prediction. Zhao et al. [25] and Dupont [26] employ the ELBO decomposition to perform feature selection by bounding mutual information. While these techniques have been very successful on many benchmark datasets (e.g., MNIST, SVHN), VAE’s semi-supervised performance on several common datasets with a lot of variation, like CIFAR-10 and CIFAR-100, was subpar. To improve the inference accuracy, Feng et al. [27] integrates the classification loss into ELBO and performs optimal interpolation (OT-approximation) in data space and latent space to overcome the “ELBO bottleneck”. However, these methods overlook the causal perspective, despite its strong relevance to SSL.

III. METHOD

This section presents the causal interpretation of our method, along with its architecture, implementation, and theoretical foundations.

Notation: The random variables are denoted by capital letters (e.g., X), while their realizations are indicated by the corresponding lowercase letters (e.g., x). Boldface letters, such as $\mathbf{X}$ and $\mathbf{x}$, represent sets of variables and their associated realized values, respectively. We use $p(X)$ to refer to the distribution or probability law associated with the random variable X in a symbolic or abstract sense and $p(x)$ to denote

the value of the density (or probability mass) at the specific realization x . Similarly, we use uppercase arguments (e.g., $MSE(\mathbf{X}; \mathbf{Y})$, $CE(X; Y)$), where CE is the cross-entropy loss and MSE is the mean squared error) to denote expected losses computed over the true data distribution, and lowercase arguments ($MSE(\mathbf{x}; \mathbf{y})$, $CE(x; y)$) to denote empirical losses computed over finite samples.

A. Preliminaries

First, we provide a concise causal rationale motivating the semi-supervised learning problem. Fig. 1(a) depicts the causal generative process considered throughout this paper. We presume that a real-world observation X (e.g., an image of a dog in a particular background) is caused by collections of generative attributes: content-related causes Z_c (e.g., texture, shape), nuisance factors Z_{nc} (e.g., background, viewpoint), and a label Y (e.g., dog). Additionally, the target Y is directly influenced by Z_c , $Z_c \rightarrow Y$, and not by Z_{nc} , $Z_{nc} \nrightarrow Y$. Moreover, we assume the content variable Z_c is nuisance-independent, i.e., $Z_c \perp\!\!\!\perp Z_{nc}$. Such causal generative mechanisms have already been taken into account [2], [28], [29], [30]. Due to the SSL setting, X is observed and Y is partially observed. Additionally, Z_c and Z_{nc} are hidden variables, making inference more difficult. In this paper, we assume only the content variable is relevant for the task. Occasionally, certain nuisance attributes may possess strong predictive power, yet relying on them would result in unreliable predictions. Consider, for example, attempting to classify airplanes based on the background of the sky. Taking the causal relationships into account for model design, we present a **Semi-supervised Model for classification using CAusal Variational Autoencoder (SEMI-CAVA)**.

B. SEMI-CAVA

Supervised learning uses batches of training data in the form of input-label pairs $(\mathbf{X}_L, \mathbf{Y}_L) = \{(X_1^l, Y_1^l), \dots, (X_{N_l}^l, Y_{N_l}^l)\}$ with the i -th observation $X_i^l \in \mathbf{R}^d$ and the corresponding class label $Y_i^l \in \{1, 2, \dots, k\}$, which are drawn from a labeled dataset $\mathbb{D}_L$. In semi-supervised learning, we try to leverage an additional set of unlabeled data $\mathbf{X}_U = \{X_1^u, \dots, X_{N_u}^u\}$ with $X_i^u \in \mathbf{R}^d$ sampled from an unlabeled dataset $\mathbb{D}_U$ in order to get a more accurate model than if we had used only the labeled dataset $\mathbb{D}_L$. We do not have access to the labels Y_i^u . When it is evident that we are referring to terms associated with a single data point, we will omit the index i . For brevity, we will omit the superscripts u and l when discussing statements that hold for both labeled and unlabeled data points. We now design a deep causal variational autoencoder for semi-supervised learning (SEMI-CAVA) that exploits the underlying causal description (in the form of a directed acyclic graph) of the data-generating process. Fig. 1(b) provides an overview of our method.

1) *Generation:* Delineating causal generative descriptions of the data, the causal graph in Fig. 1(a) returns a causal Markov factorized model:

$$p_\theta(x, z_c, z_{nc}, y) = p_{\tilde{\theta}}(x|z_c, z_{nc}, y)p_\psi(y|z_c)p(z_c)p(z_{nc}) \quad (1)$$

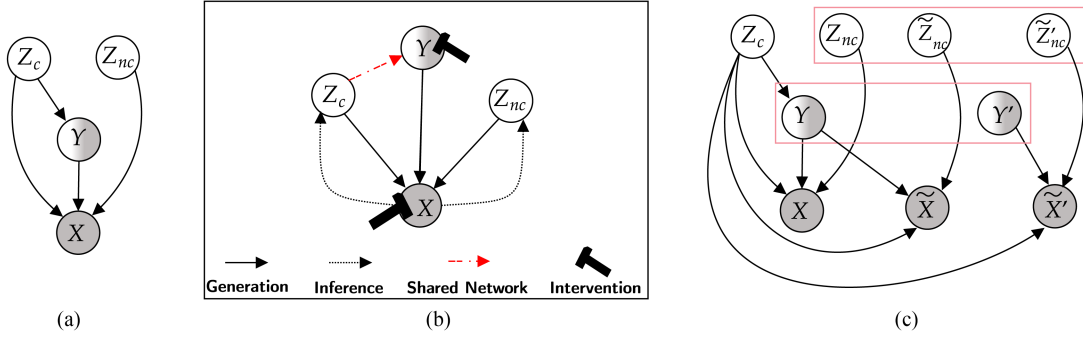


Fig. 1. (a) The causal graph for semi-supervised image classification, where grey nodes (image X) denote observable variables and white nodes (content-related causes Z_c , nuisance factors Z_{nc}) unobservable ones. Label Y is partially observed—hence, its node is half grey and half white. (b) Graphical presentation of SEMI-CAVA. Solid arrows represent the generative module following the causal structure, while dotted arrows indicate the inference module. The red arrow is involved in both inference and generation. We apply stochastic interventions on X and Y . (c) An interventonal view of mixup combined with weak augmentation. Tilded variables (e.g., $\tilde{X}$) are weakly augmented; primed variables (e.g., Y') are mix-up versions.

where the generation process is parameterized by $\theta = \{\tilde{\theta}, \psi\}$. The generation process further makes the following parametric assumptions about the conditional distributions of X and Y given their parents:

$$p_{\tilde{\theta}}(x|z_c, z_{nc}, y) = \mathcal{N}(x; f_{\tilde{\theta}}(z_c, z_{nc}, y), \sigma^2 \mathbf{I}) \quad (2)$$

$$p_{\psi}(y|z_c) = \text{Cat}(y|g_{\psi}(z_c)) \quad (3)$$

where $\text{Cat}(y|\cdot)$ is a multinomial distribution of the label, $f_{\tilde{\theta}}$ is the decoder neural network, g_{ψ} is a deep generative classifier that outputs a probability vector, and $\mathcal{N}$ represents gaussian distribution.

2) *Inference*: We use amortization [31], [32] to design an inference network for effective posterior inference:

$$q_{\phi}(z_c, z_{nc}, y|x) = q_{\phi}(y|z_c, z_{nc}, x) q_{\phi}(z_c, z_{nc}|x), \quad (4)$$

where ϕ is the collection of variational parameters. In order for the model to learn disentangled representations, during inference, we assume Z_c and Z_{nc} are conditionally independent given X , i.e.,

$$q_{\phi}(z_c, z_{nc}|x) = q_{\phi_c}(z_c|x) q_{\phi_{nc}}(z_{nc}|x). \quad (5)$$

We assume that Z_{nc} and X do not contain further information about Y given its parent Z_c . As a consequence, Y is independent of Z_{nc} and X given Z_c during inference, despite the fact that it is not implied by the causal graphical model. Therefore, we infer Y extracting information only from Z_c , which is naturally desirable. Such assumptions are made in prior work (Zhang et al. [6]).

Moreover, this allows us to share the classifier g_{ψ} between the generation process and the inference module as

$$q_{\phi}(y|z_c, z_{nc}, x) = q_{\phi}(y|z_c) = q_{\psi}(y|z_c). \quad (6)$$

The inference process also assumes the posterior distribution of Z_c and Z_{nc} given X as

$$q_{\phi_c}(z_c|x) = \mathcal{N}(z_c; \mu_{\phi_c}(x), \sigma_{\phi_c}^2(x)), \quad (7)$$

$$q_{\phi_{nc}}(z_{nc}|x) = \mathcal{N}(z_{nc}; \mu_{\phi_{nc}}(x), \sigma_{\phi_{nc}}^2(x)), \quad (8)$$

where $\tilde{\phi} = \{\phi_c, \phi_{nc}\}$ is the parameters of the encoder network. The posterior distribution of Y given Z_c as

$$q_{\psi}(y|z_c) = \text{Cat}(y|g_{\psi}(z_c)), \quad (9)$$

where the functions μ_{ϕ_c} , $\sigma_{\phi_c}^2$, $\mu_{\phi_{nc}}$, and $\sigma_{\phi_{nc}}^2$ are all represented as deep neural networks. Nota bene: $\phi = \{\tilde{\phi}, \psi\}$. With abuse of notation, we write $e(x) = (\mu_{\phi_c}(x), \sigma_{\phi_c}^2(x), \mu_{\phi_{nc}}(x), \sigma_{\phi_{nc}}^2(x))$.

3) *Prediction*: We are interested in utilizing the distribution $q_{\psi}(y|z_c)$, parameterized by g_{ψ} , for constructing the most effective classifier feasible as our inference model. Therefore, we manoeuvre $q_{\psi}(y|z_c)$ as a surrogate for $q_{\psi}(y|x)$, i.e.,

$$q_{\psi}(y|x) = q_{\psi}(y|z_c). \quad (10)$$

Moreover, we make the inference module more confident about its prediction by sharpening [11] $q_{\psi}(y|z_c)$. The sharpening parameter T is a hyperparameter (not learnable) in the model. The distribution $q_{\psi}(y|z_c)$ is also utilized during testing for the prediction of any unobserved data.

C. Training Objective

For the SSL paradigm, we must consider two different scenarios to explore in a variational framework [22]. For the unlabeled data $X^u \sim \mathbb{D}_U$, label Y^u is treated as a latent variable over which, along with Z_c and Z_{nc} , we carry out posterior inference and the model maximizes the evidential lower bound (ELBO) of $\log p(X^u)$ as (derived in Appendix A, available online)

$$\begin{aligned} \log p_{\theta}(X^u) &\geq \mathbb{E}_{q_{\phi}(Z_c, Z_{nc}, Y^u|X^u)} (\log p_{\tilde{\theta}}(X^u|Z_c, Z_{nc}, Y^u)) \\ &\quad - D_{KL}(q_{\phi_c}(Z_c|X^u)||p(Z_c)) \\ &\quad - D_{KL}(q_{\phi_{nc}}(Z_{nc}|X^u)||p(Z_{nc})) \\ &\quad - D_{KL}(q_{\psi}(Y^u|Z_c)||p(Y)) \\ &= \text{ELBO}_{\mathbb{D}_U}(X^u), \end{aligned} \quad (11)$$

where $p(Y)$ is the true marginal distribution for labels Y which may be approximated (in most cases) using $\hat{p}(Y^l|X^l)$ in the labeled dataset. Therefore, KL divergence between $q_{\psi}(Y^u|Z_c)$ and $p(Y)$ impose a constrain that encourages the class predictions to be more aligned with the true class distribution. For the

labeled data $(X^l, Y^l) \sim \mathbb{D}_L$, the label related to the datapoint is observed and the corresponding ELBO is constructed as (derived in Appendix A, available online)

$$\begin{aligned} & \log p_\theta(X^l, Y^l) \\ & \geq \mathbb{E}_{q_{\phi_c}(Z_c|X^l)q_{\phi_{nc}}(Z_{nc}|X^l)\tilde{p}(Y^l|X^l)} (\log p_{\tilde{\theta}}(X^l|Z_c, Z_{nc}, Y^l)) \\ & \quad + \log(q_\psi(Y^l|X^l)) - D_{KL}(q_{\phi_c}(Z_c|X^l)||p(Z_c)) \\ & \quad - D_{KL}(q_{\phi_{nc}}(Z_{nc}|X^l)||p(Z_{nc})) \\ & = \text{ELBO}_{\mathbb{D}_L}(X^l, Y^l) \end{aligned} \quad (12)$$

where $\tilde{p}(Y^l|X^l)$ is empirically degenerated distribution, i.e., $\tilde{p}(y^l|x^l) = 1$ for all label-input pairs (x^l, y^l) . As $\log q_\psi(Y^l|X^l)$ naturally arrives at $\text{ELBO}_{\mathbb{D}_L}$, unlike [27] and [22], we do not use additional classification loss or label smoothing [33] to integrate classification loss into the ELBO.

1) *Interventional Consistency*: One fundamental assumption we have made is that the content Z_c is independent of the nuisance factors Z_{nc} . However, this presumption may not always be the case. There may exist a spurious correlation and we must implement measures to eliminate that. Data augmentation may be thought of as a collection of interventions that eliminates spurious correlations between content features and nuisance factors [34], [35], [36]. However, in a VAE, the encoder tends to map the same image and its semantics-preserving transformation (augmentation) to different locations in the latent space. This encoding “inconsistency” degrades learned representations, which impacts generalization [37]. To mitigate this issue, we think of mixup augmentation [9] as a stochastic intervention [38] on the causal VAE system and we define consistency losses to regularize the latent space. For the labeled data, we intervene on the observed variables X^l and Y^l (at the batch level) as follows:

$$\begin{aligned} \begin{pmatrix} \mathbf{X}_L \\ \mathbf{Y}_L \end{pmatrix} & \leftarrow \lambda \begin{pmatrix} \mathbf{X}_L \\ \mathbf{Y}_L \end{pmatrix} + (1 - \lambda) \begin{pmatrix} \delta_{\mathbf{X}_L} \\ \delta_{\mathbf{Y}_L} \end{pmatrix}, \\ \begin{pmatrix} \delta_{\mathbf{X}_L} \\ \delta_{\mathbf{Y}_L} \end{pmatrix} & \sim \mathbb{D}_L, \lambda \sim \text{Beta}(\alpha_L, \beta_L). \end{aligned} \quad (13)$$

For the unlabeled data, because Y^u is considered as latent, we intervene only on X^u :

$$\mathbf{X}_U \leftarrow \lambda \mathbf{X}_U + (1 - \lambda) \delta_{\mathbf{X}_U}, \quad \delta_{\mathbf{X}_U} \sim \mathbb{D}_U, \lambda \sim \text{Beta}(\alpha_U, \beta_U). \quad (14)$$

(α_L, β_L) and (α_U, β_U) are the shape parameters of the Beta distributions, controlling the strength and symmetry of sample mixing for the labeled and unlabeled datasets, respectively. Let us define the convenient short-hand notation: $\lambda(a, b) := \lambda a + (1 - \lambda)b$. In order to make the latent space consistent, we define the following interventional consistency losses (ICL): For interventions on labeled data,

$$\begin{aligned} & \text{ICL}_{\mathbb{D}_L}(\mathbf{X}_L, \mathbf{Y}_L, \lambda) \\ & = CE\left(g_\psi\left(\mathbf{Z}_c^{\lambda(\mathbf{X}_L, \delta_{\mathbf{X}_L})}\right); \lambda(\mathbf{Y}_L, \delta_{\mathbf{Y}_L})\right) \\ & \quad + MSE(e(\lambda(\mathbf{X}_L, \delta_{\mathbf{X}_L})); \lambda(e(\mathbf{X}_L), e(\delta_{\mathbf{X}_L}))), \end{aligned} \quad (15)$$

Algorithm 1: Training Process of SEMI-CAVA at t -th Iteration.

Input:Labeled batch $(\mathbf{x}_L, \mathbf{y}_L)$ of batch size b_L Unlabeled batch $\mathbf{x}_U$ of batch size b_U

Mixup parameters:

$$\lambda_l \sim \text{Beta}(\alpha_L, \alpha_L + 1), \quad \lambda_u \sim \text{Beta}(\alpha_U, \alpha_U + 1)$$

Loss weight w_t Model's parameters: $\Theta_{t-1} = (\tilde{\theta}_{(t-1)}, \psi_{(t-1)}, \tilde{\phi}_{(t-1)})$

Model optimizer: SGD

1:

$$L_{\mathbb{D}_L} = \frac{-w_t}{b_L} \sum_{\substack{x_i^l \in \mathbf{x}_L \\ y_i^l \in \mathbf{y}_L}} \text{ELBO}_{\mathbb{D}_L}(x_i^l, y_i^l) + \text{ICL}_{\mathbb{D}_L}(\mathbf{x}_L, \mathbf{y}_L, \lambda_l)$$

2:

$$L_{\mathbb{D}_U} = \frac{-w_t}{b_U} \sum_{x_i^u \in \mathbf{x}_U} \text{ELBO}_{\mathbb{D}_U}(x_i^u) + \text{ICL}_{\mathbb{D}_U}(\mathbf{x}_U, \lambda_u)$$

3: $L = L_{\mathbb{D}_L} + L_{\mathbb{D}_U}$ 4: $\Theta_t = \text{SGD}\left(\Theta_{t-1}, \frac{\partial L}{\partial \Theta_{t-1}}\right)$ **Output:**Updated parameters: Θ_t

where Z_c^t is sampled from $\mathcal{N}(\mu_c(t); \sigma_c^2(t))$. Similarly, for interventions on unlabeled data, we define

$$\begin{aligned} & \text{ICL}_{\mathbb{D}_U}(\mathbf{X}_U, \lambda) \\ & = CE\left(g_\psi\left(\mathbf{Z}_c^{\lambda(\mathbf{X}_U, \delta_{\mathbf{X}_U})}\right); \lambda\left(g_\psi\left(\mathbf{Z}_c^{\mathbf{x}_U}\right), g_\psi\left(\mathbf{Z}_c^{\delta_{\mathbf{x}_U}}\right)\right)\right) \\ & \quad + MSE(e(\lambda(\mathbf{X}_U, \delta_{\mathbf{X}_U})); \lambda(e(\mathbf{X}_U), e(\delta_{\mathbf{X}_U}))). \end{aligned} \quad (16)$$

However, unlike [27], we do not perform any optimal interpolation in latent space. Approximating optimal interpolation could be expensive in many settings. The primary goal of this work is not to learn disentanglement representation. However, in the Section V, under mild assumptions, we have shown that interventional consistency loss with proper data augmentation encourages disentanglement representation.

D. Algorithm

E. SEMI-CAVA Pipeline

Fig. 2 presents a high-level overview of SEMI-CAVA: a batch of images $\mathbf{x}$ (standard augmented or mixup applied on top of those augmentations — both variants are denoted simply as $\mathbf{x}$ in the diagram for visual economy) is processed by a shared convolutional encoder; its output then passes through two parallel MLP heads that emit the means and variances of two disentangled Gaussian posteriors from which we sample $\mathbf{z}_c$ (causal factors) and $\mathbf{z}_{nc}$ (nuisance factors). The causal factors $\mathbf{z}_c$ are fed to a classifier g_ψ that outputs a probability vector and thereby produces a label prediction $\hat{y}$. (when ground-truth labels are available we use them to nudge the network toward

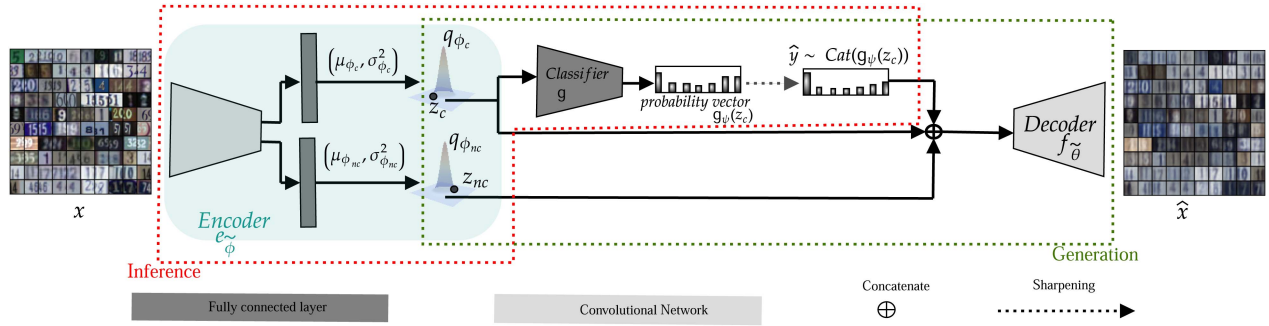


Fig. 2. The schematic of SEMI-CAVA. The latent representation Z is divided into two parts: causal (or content) representation Z_c and nuisance (or style) representation Z_{nc} . While only content representations participate in classification, both content and style representations, along with class labels, contribute to the reconstruction process.

correct predictions, whereas for unlabeled images, $\hat{y}$ is treated as a pseudo-label for training). This same label code $\hat{y}$ concatenated with $z = (z_c, z_{nc})$ is fed into the decoder $f_{\tilde{\theta}}$ (to reconstruct the image). Training minimizes a composite objective that combines both a variational ELBO and a consistency loss, each designed separately for the labeled and unlabeled cases, as detailed in Sections III-C and Algorithm 1. While the weakly augmented inputs are used for ELBO computation, the mixup-applied inputs serve solely in computing interventional-consistency losses, and are not reconstructed. The implementation specifics are presented in Appendix C, available online.

IV. RESULTS

We compare SEMI-CAVA against state-of-the-art semi-supervised learning methods on standard benchmark datasets: the CIFAR-10 and CIFAR-100 datasets [39], and the SVHN dataset [40]. Additionally, we carried out a comprehensive and systematic series of experiments to assess the SEMI-CAVA in terms of its scalability to diverse medical image modalities, including blood cell microscopy, breast ultrasound, dermatoscope, chest x-ray, and abdominal CT. Notably, our approach consistently outperformed current state-of-the-art semi-supervised deep-learning benchmarks in medical image classification.

A. MedMNIST Experiment

We consider lightweight MedMNIST datasets [41], which are well-suited for the rapid prototyping of new algorithms. We assessed the disparity between expert annotations and SEMI-CAVA-generated classifications using two evaluation metrics: Accuracy (ACC), expressed in percentage scale, and Area under the ROC (receiver operating characteristic) Curve (AUC). AUC typically ranges in value from 0 to 1. However, to ensure a fair comparison with the MedMNIST benchmark [41], we have rescaled the AUC values by a multiplying factor of 100. AUC is less sensitive to class imbalance than ACC. The MedMNIST benchmark poses challenges for three main reasons:

- 1) It is a fully supervised benchmark.
- 2) It includes larger models such as ResNet 18 (11 M) and ResNet 50 (23 M).
- 3) The benchmark also includes the results obtained from the utilization of high-resolution images (224×224 pixels).

For the purpose of comparison, we consider only the top three results from the MedMNIST benchmark for each individual dataset. The results are summarized in Table I.

1) *Organ{A,C,S}MNIST*: Precise organ classification is pivotal in medical image analysis, serving as the foundation for subsequent tasks such as organ segmentation, localization, and image registration. Organ{A,C,S}MNIST datasets are derived from 3D computed tomography (CT) scans obtained from the Liver Tumor Segmentation Benchmark (LiTS) [42], differing primarily in anatomical viewpoints they represent, specifically axial, coronal, and sagittal planes respectively for OrganA, OrganC, and OrganS datasets. The SEMI-CAVA model, trained with only 10% labeled data, outperforms the fully supervised MedMNIST baseline in terms of accuracy for both OrganCMNIST and OrganSMNIST datasets. For OrganAMNIST, the SEMI-CAVA model achieves an accuracy slightly lower by 0.9 and 0.5 percentage points compared to the highest and second-highest accuracies of the MedMNIST benchmark, respectively. However, the top two models in the benchmark utilize high-resolution 224×224 images, whereas SEMI-CAVA excels even with lower resolution 28×28 images. SEMI-CAVA, at the annotation ratio of 0.1, surpasses all other models that use 28×28 image size as inputs. As most models, including SEMI-CAVA, consistently yield nearly identical AUC scores, it highlights that AUC may not serve as a decisive metric in this particular case.

For the rest of the MedMNIST datasets, we compare our results with the MedMNIST benchmark as well as the PL and DePL models [43] which use the same MedMNIST dataset and perform a semi-supervised classification with an annotation ratio of 0.1.

2) *BloodMNIST*: BloodMNIST is a dataset comprising microscopic peripheral blood cell images (organized into eight classes) obtained from individuals without infection, hematologic or oncologic diseases, and free from pharmacologic treatment at the time of blood collection [44]. The experiment shows SEMI-CAVA significantly improves over PL and DePL by a margin of 12.66% in accuracy and 1.94 points in AUC. Interestingly, it even exceeds the headroom supervised learning on the full dataset (MedMNIST benchmark).

3) *PneumoniaMNIST*: The PneumoniaMNIST dataset consists of MNIST-like pediatric chest X-ray images [45], with

TABLE I
COMPARISON OF ACCURACY (ACC) AND AREA UNDER CURVE (AUC) SCORES FOR CLASSIFICATION PERFORMANCE ON THE MEDMNIST DATASET

Datasets	Methods	Annotation Ratio	Accuracy	AUC
OrganAMNIST	MedMNIST Benchmark	1	95.1*	99.8*
			94.7*	99.8*
			93.5	99.7
	SEMI-CAVA	0.1	94.21 $\pm$ 0.27	99.54 $\pm$ 0.12
		0.25	94.88 $\pm$ 0.2	99.72 $\pm$ 0.01
OrganCMNIST	MedMNIST Benchmark	1	92.0*	99.4*
			91.1*	99.3*
			90.5	99.2
	SEMI-CAVA	0.1	92.35 $\pm$ 0.30	99.14 $\pm$ 0.08
		0.25	93.47 $\pm$ 0.29	99.2 $\pm$ 0.04
OrganSMNIST	MedMNIST Benchmark	1	81.3	97.4
			78.5*	97.5*
			78.2	97.2
	SEMI-CAVA	0.1	81.62 $\pm$ 0.1	97.15 $\pm$ 0.18
		0.25	83.55 $\pm$ 0.1	97.77 $\pm$ 0.09
BloodMNIST	MedMNIST Benchmark	1	96.6	99.8
			96.3*	99.8*
			95.8	99.8
	PL	0.1	84.09 $\pm$ 1.17	97.83 $\pm$ 0.23
	DePL	0.1	83.68 $\pm$ 0.59	97.72 $\pm$ 0.15
	SEMI-CAVA	0.1	96.75 $\pm$ 0.09	99.77 $\pm$ 0.01
PneumoniaMNIST	MedMNIST Benchmark	1	94.6	99.1
			88.4*	96.2*
			87.8	94.7
	PL	0.1	85.83 $\pm$ 2.13	94.34 $\pm$ 0.91
	DePL	0.1	84.36 $\pm$ 3.79	94.15 $\pm$ 0.33
	SEMI-CAVA	0.1	92.68 $\pm$ 1.35	95.19 $\pm$ 0.54
DermaMNIST	MedMNIST Benchmark	1	76.8	91.4
			75.4*	92.0*
			74.9	91.5
	PL	0.1	68.88 $\pm$ 1.03	82.64 $\pm$ 1.19
	DePL	0.1	69.30 $\pm$ 0.85	83.82 $\pm$ 0.95
	SEMI-CAVA	0.1	72.28 $\pm$ 0.32	86.33 $\pm$ 1.9
		0.3	75.79 $\pm$ 0.63	91.57 $\pm$ 0.53
BreastMNIST	MedMNIST Benchmark	1	86.1	91.9
			86.3	90.1
			84.2*	86.6*
	PL	0.1	74.74 $\pm$ 1.04	74.67 $\pm$ 3.68
	DePL	0.1	76.67 $\pm$ 1.32	75.33 $\pm$ 3.05
	SEMI-CAVA	0.1	83.33 $\pm$ 3.20	87.23 $\pm$ 1.07

Asterisks (*) denote models trained using 224 $\times$ 224 images.

the primary objective being binary-class classification, distinguishing between pneumonia and normal cases. SEMI-CAVA surpasses both PL and DePL by achieving a top-1 accuracy increase of over 6%, rising from 85.83% to 92.68%. Additionally, it enhances the AUC from 94.34 to 95.19, representing a significant improvement margin.

4) *DermaMNIST*: The DermaMNIST dataset is based on the HAM10000 dataset [46], [47], which contains a diverse set of dermatoscopic images depicting various pigmented skin lesions. These images have been categorized into seven distinct disease types. The SEMI-CAVA, with an annotation ratio of 0.1, exhibits better performance than the DePL model, achieving a 2.98% higher accuracy and a 2.51-point increase in AUC. Raising the annotation ratio to 0.3, SEMI-CAVA achieves a further 3.51% boost in accuracy and a 5.33-point rise in AUC compared to the 0.1 annotation ratio setting. This is particularly impressive as we outperform all models within the MedMNIST benchmark

in terms of both accuracy and AUC, except the top-performing one.

5) *BreastMNIST*: The BreastMNIST dataset is made up of breast ultrasound images [48] categorized into three classes: normal, benign, and malignant. Due to the utilization of low-resolution images, previous research efforts [41], [43] have simplified the task by merging normal and benign cases into a single positive category and distinguishing them from malignant cases, which are treated as negative. BreastMNIST poses a unique challenge with merely 546 training samples, comprising 493 normal and 53 malignant instances. Randomly selecting only 10% of the training data as labelled samples exacerbates the challenge by providing minimal supervision for training the model. The experimental results showcase SEMI-CAVA's significant enhancement over PL and DePL, outperforming them by at least 6.66% in accuracy and 11.90 points in AUC. Additionally, SEMI-CAVA attains an F1 score of 78.32($\pm$ 3.64). It exhibits a

TABLE II
COMPARISON OF ERROR RATES IN VISION DATASETS. MOST OF THE NUMBERS ARE TAKEN FROM PREVIOUS PAPERS

Method	CIFAR-10		CIFAR-100		SVHN
	250	4K	4K	10K	1K
VAT	36.03±2.82	11.05 ±0.31	-	37.78	5.98±0.21
II-Model	53.02±2.05	14.01±0.38	-	39.19	7.54±0.36
Pseudo-Labeling (PL)	49.78±0.43	16.09±0.28	-	-	9.94±0.61
Mean Teacher	32.32±2.30	9.19±0.19	44.71	38.92	3.57±0.11
Mix-up	47.43±0.92	10.71±0.44	46.61±0.88	38.62±0.67	16.79±0.63
Mix-Match (K=1)	17.09[†]	8.06 [†]	-	35.99	-
SHOT-VAE	28.01 ± 6.17*	8.51±0.32	40.58±0.48	31.41±0.21	6.45±0.63*
SEMI-CAVA	19.81±2.05	7.32±0.31	37.57±0.15	30.90±0.06	5.92±0.17

The best number is highlighted in bold. The asterisks (*) indicate the outcomes of SHOT-VAE trained by us. The daggers (†) denote the results of MixMatch for K=1, meaning without distribution averaging.

performance level comparable to that of supervised learning on the entire dataset.

The current SEMI-CAVA framework employs the Wide-ResNet-28-2 architecture as its encoder network across all medical image modalities. Furthermore, in our MedMNIST experiments, we solely utilize images with dimensions of 28×28 pixels. However, we are of the opinion that enhancing the accuracy of SEMI-CAVA is attainable through high-resolution images with the adoption of a larger encoder network, such as ResNet-18, or even ResNet-50.

B. CIFAR-10, CIFAR-100, and SVHN Experiments

1) *Baseline Methods*: For comparison, we consider the six methods as baselines: II-model [49], Virtual Adversarial Training [10], Pseudo-Label [12], Mean Teacher [13], SHOT-VAE [27], and Mix-Match [11]. Furthermore, we consider the Mix-up [9] technique on its own as a baseline. For a fair comparison, we solely evaluate SEMI-CAVA against techniques that employ the same architectures, while refraining from comparing against methods that use larger architectures (such as “Wide ResNet-28-10”) and strong augmentation. Since it is generally accepted that more powerful architectures and strong augmentation may boost the performance of any approach, including SEMI-CAVA (see Appendix D-A, available online), we impose these constraints on our baselines. We compare our results to those reported by earlier research works [11], [27], [50], [51].

2) *Results*: Table II displays the outcomes obtained using SEMI-CAVA, juxtaposed with results from other methods. The results indicate that, in a reasonably fair comparison, SEMI-CAVA significantly improves over SHOT-VAE, which currently stands as the leading semi-supervised VAE model. Compared to results reported in past papers, SEMI-CAVA has achieved the lowest error rates among the same model architectures on three benchmarks: CIFAR-10-4 K, CIFAR-10-4 K, and CIFAR-100-10 K. However, for CIFAR-10-250, it achieves an 18.95% error rate with 250 labels, closely rivaling Mix-Match, and for SVHN-1 K, SEMI-CAVA attains a 5.65% error rate, approaching the performance level of the leading approach, Mean Teacher.

To the best of our knowledge, although superior results on these datasets have been achieved, they typically require the utilization of larger models, strong augmentation techniques,

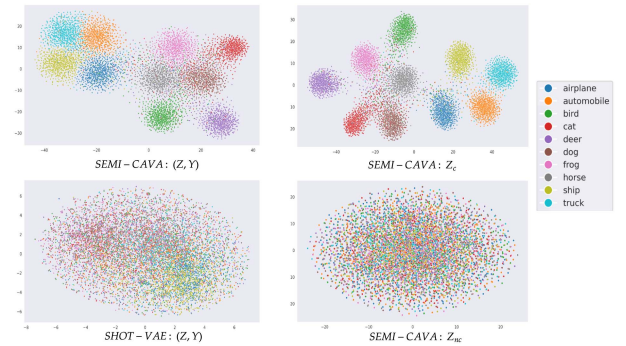


Fig. 3. **Left column**: t-SNE plots of $(z, y) \sim q$ given the test dataset for SEMI-CAVA and SHOT-VAE. **Right column**: t-SNE plots of $z_c \sim q_{\phi_c}$ and $z_{nc} \sim q_{\phi_{nc}}$ given the test dataset for SEMI-CAVA. Note that, in case of SEMI-CAVA, we have $z = (z_c, z_{nc})$.

more robust regularization techniques, or additional distillation procedures.

3) *Ablation Study*: We analyse the t-SNE plots of latent representations learned through SEMI-CAVA and SHOT-VAE. Other baseline classification models do not possess latent space. As can be discerned from the Fig. 3, the representations produced by the SEMI-CAVA exhibit notably higher coherence compared to those generated by the SHOT-VAE. Moreover, this coherence is attributed to the causal dimensions (z_c) within the latent representation.

Equipped with the consistent latent space, our objective now shifts to investigate: are the causal representations invariant? The anticipated outcome is that these causal representations should exhibit invariance as they are utilized in predicting class labels. To explore this, we examine the tsne plots of z_c 's obtained by applying various data augmentation (intervention) schemes to the test set. Taking representations under null intervention (raw images) as the baseline, we qualitatively measure the invariance by overlaying the representations of perturbed images. The degree of overlap between these representations serves as a visual indicator of the model's capacity to extract invariant features, which are pivotal for generalization. In Fig. 4, we observe that under the same training augmentation scheme, causal representations generally remain invariant across classes, with the exception of deer, frog, and cat classes. During the training phase, we use zero padding, random cropping, and a random horizontal flip with a probability of 0.5. However,

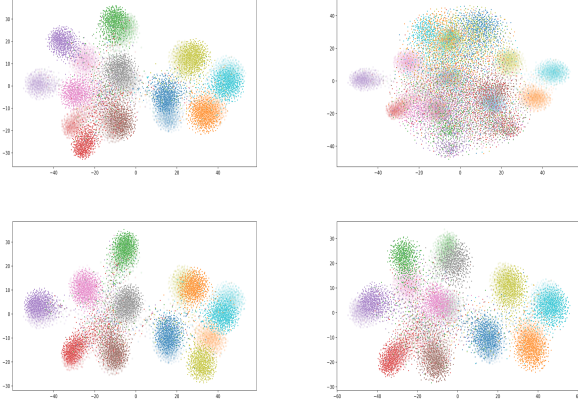


Fig. 4. Qualitative evaluation of the invariance property of causal dimensions (z_c) given the test dataset. The baseline (null intervention) scatterplot is rendered with transparent circle markers, whereas representations under different augmentation schemes are depicted with solid dot markers. **Top left:** train augmentation vs. baseline. **Top right:** vertical flip vs. baseline. **Bottom left:** horizontal flip vs. baseline. **Bottom right:** grayscale vs. baseline.

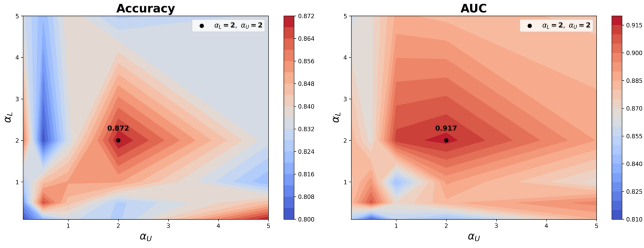


Fig. 5. Contour plots of Accuracy (left) and AUC (right) as a function of α_L and α_U . The black dot marks the optimal configuration (2,2).

when solely applying horizontal flipping with a probability of 1, the invariance holds for all classes except the automobile and the ship. A vertical flip with a probability of 1 completely disrupts the coherence of the latent space. On the other hand, when applying grayscale transformation, causal representations generally maintain invariance across classes, except for the bird, frog, and horse classes.

4) *Hyperparameter Study:* In general, there are four hyperparameters ($\alpha_L, \beta_L, \alpha_U, \beta_U$) governing the Beta distributions (in Algorithm 1), which makes a comprehensive hyperparameter study complex and computationally intensive. However, under our assumption from Theorem 2 — namely, $(\beta = \alpha + 1)$ — the search space becomes significantly more manageable. To investigate the sensitivity of the Beta distribution parameters used in our mixup augmentation strategy, we perform a grid-based hyperparameter sweep over $\alpha_L, \alpha_U \in \{0.1, 0.5, 1.0, 2.0, 5.0\}$ on the BreastMNIST dataset, and visualize the results using contour plots.

Fig. 5 illustrates how our model’s performance varies across two Beta distribution parameters: α_L (vertical axis) and α_U (horizontal axis). High sensitivity is expected on the BreastMNIST dataset due to its very limited size. In both contour plots, red regions indicate higher performance, while blue regions represent lower performance. A distinct peak is observed at $\alpha_L = 2$ and $\alpha_U = 2$, marked in both plots by a black dot. Although performance gradually declines as we move away

from this region, our method consistently outperforms the PL and DePL baselines across all hyperparameter combinations evaluated.

V. THEORETICAL INSIGHTS

Our objective is to demonstrate the capability of identifying content representations under a minimal set of assumptions. Notably, our goal is not to isolate individual latent factors. Rather, we aim to separate content from style such that content variables can be effectively utilized for subsequent classification task.

A. Data Generation

$$X = f_X(Y, Z_c, Z_{nc}), Y = f_Y(Z_c), Z_c \sim p_{Z_c}, Z_{nc} \sim p_{Z_{nc}} \quad (17)$$

We assume f_X restricted to $Z = (Z_c, Z_{nc})$, i.e., $f_X|_Z : \mathcal{Z} \rightarrow \mathcal{X}$ to be diffeomorphism between open, simply-connected n -dim spaces $\mathcal{Z}$ and $\mathcal{X}$. Note that, the invertibility of $f_X|_Z$ implies that, despite X being high-dimensional with $d \geq \dim(Z)$, $\mathcal{X}$ can be seen as $\dim(Z)$ dimensional sub-manifold embedded in $\mathbb{R}^d$. We further assume an augmentation, represented as $\tilde{X}$, is derived from the original input X by subjecting a stochastically modified representation of Z , i.e. $\tilde{Z}$, to the same structural assignment f . We conceptualize the augmented representation $\tilde{z} \in \mathcal{Z}$ as the result of applying $f_X|_Z^{-1}$ to an augmented observation $\tilde{x} = t(x) \in \mathcal{X}$. Similarly, mixup representation $\tilde{z}' \in \mathcal{Z}$ can be understood as the outcome of employing the inverse function $f_X|_Z^{-1}$ on the mixup observation $\tilde{x}' = \Lambda(\tilde{x}) = \Lambda(t(x)) \in \mathcal{X}$.

Kügelgen et al. [35] incorporate the concept that the set of transformations $\mathcal{T}$ utilized for augmentation commonly adheres to the principle that each transformation $t \in \mathcal{T}$ preserves specific facets (content) of the data. To this end, we further posit that the mixup operation also preserves the content of the data. The process of generating an augmented mixup observation $\tilde{X}'$ is thus described by (see Fig. 1(c))

$$Z \sim p_Z, X = f_X|_Z(Z), \quad (18)$$

$$\tilde{Z}|Z \sim p_{\tilde{Z}|Z}, \tilde{X} = f_X|_Z(\tilde{Z}), \quad (19)$$

$$\tilde{Z}'|\tilde{Z} \sim p_{\tilde{Z}'|\tilde{Z}}, \tilde{X}' = f_X|_Z(\tilde{Z}'). \quad (20)$$

Without loss of generality (WLOG), we can assume that Z_c corresponds to the initial $\dim(Z_c)$ dimensions of Z .

Assumption 1: (Content-invariance under mixup and augmentation)

$$p_{\tilde{Z}'|\tilde{Z}}(\tilde{z}'|\tilde{z}) = \delta(\tilde{z}' - \tilde{z}_c) \times p_{\tilde{Z}'_{nc}|\tilde{Z}_{nc}}(\tilde{z}'_{nc}|\tilde{z}_{nc})$$

$$p_{\tilde{Z}|Z}(\tilde{z}|z) = \delta(\tilde{z}_c - z_c) \times p_{\tilde{Z}_{nc}|Z_{nc}}(\tilde{z}_{nc}|z_{nc})$$

for continuous densities $p_{\tilde{Z}'_{nc}|\tilde{Z}_{nc}}, p_{\tilde{Z}_{nc}|Z_{nc}}$ on $\mathcal{Z}_{nc} \times \mathcal{Z}_{nc}$, where $\delta(\cdot)$ is Dirac-delta function.

B. Identifiability

Definition 1 (Content-identifiability [35]): True content $z_c = f_X|_Z^{-1}(x)_{1:\dim(Z_c)} = f_X|_Z^{-1}(\tilde{x})_{1:\dim(Z_c)}$ is identifiable by

a function $g : \mathcal{X} \rightarrow \mathcal{Z}$ if there exists a bijection $h : \mathcal{Z}_c \rightarrow \mathcal{Z}_c$ between the inferred content $\tilde{z}_c = g(x)_{1:\dim(\mathcal{Z}_c)}$ and z_c . In other words, inferred content encompasses all and only information about z_c .

Theorem 1 (Content-identifiability under a generative model [35]): Consider the process of generating data as outlined in Section V-A, i.e., the pairs of augmented and mixup views are generated according to (19) and (20) with $p_{\tilde{Z}|\tilde{Z}}$ as defined in Assumption 1. Furthermore, let us assume that

- i) $f_{X|Z}$ is a diffeomorphism.
- ii) p_Z is a smooth, continuous density on Z with $p_Z(z) > 0$ almost everywhere.
- iii) $p_{\tilde{Z}_{nc}|\tilde{Z}_{nc}}$ is smooth w.r.t both $\tilde{z}'_{nc}$ and $\tilde{z}_{nc}$; and for any $\tilde{z}_{nc}, p_{\tilde{Z}_{nc}|\tilde{Z}_{nc}}(\cdot|\tilde{z}_{nc}) > 0$ in some open, non-empty subset containing $\tilde{z}_{nc}$.

If a generative model $(\hat{p}_{\tilde{Z}}, \hat{p}_{\tilde{Z}'_{nc}|\tilde{Z}_{nc}}, \hat{f}^{-1})$ assumes the same generative process (Section V-A) satisfying the above assumptions (i)–(iii), and matches the data likelihood,

$$p_{\tilde{X}, \tilde{X}'}(\tilde{x}, \tilde{x}') = \hat{p}_{\tilde{X}, \tilde{X}'}(\tilde{x}, \tilde{x}') \quad \forall (\tilde{x}, \tilde{x}') \in \mathcal{X} \times \mathcal{X},$$

then it identifies the true content variables via $g = \hat{f}^{-1}$ in the sense of Definition 1.

Proof: See Theorem 4.2 in [35] or Appendix B, available online. $\square$

Theorem 2 (Identifying content with a non-invertible encoder): Under the data generation process as outlined in Section V-A, along with the conditions (i)–(iii) as detailed in Theorem 1, let $g : \mathcal{X} \rightarrow \mathcal{Z}_c$ be any smooth function that minimizes the following functional:

$$\mathcal{L}(g) := \mathbb{E}_{(\tilde{X}', \tilde{X}) \sim p_{\tilde{X}', \tilde{X}}} \|g(\tilde{X}') - g(\tilde{X})\|_2^2 + \beta D_{KL}(g(\tilde{X}) \| p_c),$$

where $\beta > 0$, D_{KL} is the Kullback–Leibler divergence, and p_c is a regular density function on $\mathcal{Z}_c$, i.e., $0 < p_c < \infty$. Then g identifies the true content as delineated by Definition 1.

Proof: See Theorem 4.4 in [35] or Appendix B, available online. $\square$

Theorem 3: Assume $\alpha + 1 = \beta$. Then,

$$\begin{aligned} & \mathbb{E}_{\substack{\tilde{X}, \tilde{\delta}_{\tilde{X}} \sim p_{\tilde{X}} \\ \lambda \sim \text{Beta}(\alpha, \beta)}} \left\| e_{\tilde{\phi}}(\lambda(\tilde{X}, \tilde{\delta}_{\tilde{X}})) - \lambda(e_{\tilde{\phi}}(\tilde{X}), e_{\tilde{\phi}}(\tilde{\delta}_{\tilde{X}})) \right\|_2^2 \\ &= \mathbb{E}_{\substack{\tilde{X}, \tilde{\delta}_{\tilde{X}} \sim p_{\tilde{X}} \\ \lambda \sim D_{\beta}}} \left\| e_{\tilde{\phi}}(\lambda(\tilde{X}, \tilde{\delta}_{\tilde{X}})) - e_{\tilde{\phi}}(\tilde{X}) \right\|_2^2 - \mathbb{V}_{\tilde{X} \sim p_{\tilde{X}}} (e_{\tilde{\phi}}(\tilde{X})), \end{aligned}$$

where D_{β} is a uniform mixture of two Beta distributions, i.e., $(1 - \frac{\beta^2}{(2\beta-1)^2})\text{Beta}(\beta, \beta) + \frac{\beta^2}{(2\beta-1)^2}\text{Beta}(\beta+1, \beta-1)$.

For an appropriate Beta distribution of mixup coefficients, Theorem 3 decomposes the mean squared error within the interventional consistency loss into two parts: the mean squared error between mixup representation and augmented representation, and the intrinsic variance of augmented representations. The initial component, coupled with the KL divergence between content posterior $q_{\phi_c}(Z_c|X)$ and prior $p(Z_c)$ in the ELBO satisfies Theorem 2, thus promoting disentanglement of content and nuisance factors.

VI. DISCUSSION

We have provided a framework for integrating the causal graph into variational inference for semi-supervised learning using a generic causal graph. To this end, we adhere to the insights of Kügelgen et al. [35], aligning our approach (in Section V) with their findings that highlight the interpretability of data augmentation from a causal perspective. This synergy reinforces the understanding of our framework’s effectiveness. Our understanding identifies content as the component of the representation that remains consistently invariant across various views (or environments). Importantly, the content-style division is implicitly set by the chosen augmentation.

The identifiability result presented by Kügelgen et al. serves well in disentangling latent variables, particularly in the constrained setting where all latent variables associated with style information must be altered through data augmentation. However, it’s essential to note that certain augmentation schemes may selectively impact only a subset of style variables. To illustrate, consider two variables, z_1 (color) and z_2 (position), representing all nuisance (or style) variables. If an data augmentation scheme only alters z_1 (color), then z_2 (position) might be erroneously perceived as a content variable, hindering the identification of true content variables. In practical terms, it is feasible to induce changes in z_1 alone, for instance, through color distortions. Yet, for z_2 to change, a separate augmentation, such as cropping, is necessary. In essence, it is conceivable that only a subset of style variables is affected by specific augmentations in practice, necessitating a more nuanced approach for accurate identification of content variables. Nevertheless, it is crucial to emphasize the inherent difficulty of distinguishing between content and nuisance. This distinction demands a deep understanding of the particular working domain, and our choice of data augmentation strategies should be guided by a careful consideration of the underlying content. As an illustration, in the context of the organ classification problem, which includes classes like “femur-left,” “femur-right,” “kidney-left,” “kidney-right,” “lung-left,” and “lung-right,” we identify image orientation as a component of the content. Consequently, we refrain from applying rotation or flipping to organ images during the augmentation process. Thus, constructing the SEMI-CAVA model for SSL demands a thorough grasp of the causal relationships among the variables Z_c , Z_{nc} , X , and Y within the graphical model (see Appendix E, available online). This causal perspective (Fig. 1(b) and (c)) is indispensable not only for data augmentation but also for model design (Fig. 2).

Much like the findings of Kügelgen et al. [35], our theory holds in an asymptotic context and at the global optimum. But, in practical terms, we tackle a non-convex optimization problem with a finite sample size. In many cases, to achieve the desired performance, a domain-specific or problem-specific causal graph may be required. Thus, implementing our approach in niche domains (like medicine) requires further research. We do not model dependencies between Z_c and Z_{nc} . In most situations, there might be a causal link between them. Hence, modeling this dependency is promising an exciting avenue for future research. We assume that causal relationships among

variables are provided by experts in the field or causal discovery algorithms. However, the causal graph presented may not be accurate in both scenarios, resulting in poor performance. The true underlying causal graph is typically complex; however, the SEMI-CAVA graphical model simplifies this complexity by categorizing individual causes into distinct groups and representing each group as a single high-dimensional variable. While this level of abstraction is adequate for our current application, other use cases may require a more granular causal model. Exploring such refinements is an important direction for future research. Moreover, an efficient latent-space regularization strategy compatible with domain-specific strong augmentations is needed for better results.

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